

SUPPLEMENTARY RESULTS: M3C

This document contains the model-specific component and loss-function ablations for M3C, accompanying *Beyond the Majority Bin: A Methodological Framework for Balanced Automatic Pronunciation Assessment*. The cross-model comparisons and experimental reproducibility settings are reported in the paper’s supplementary material.

I. READING GUIDE

Baseline denotes the original model and training procedure. The +CV configuration introduces speaker-independent grouped cross-validation using `GroupKFold(n_splits=5)` without shuffling. Both +DB and +MS build on CV: +DB adds Dynamic Balanced MSE, whereas +MS adds Macro-MSE-based checkpoint selection. Complete BOMA combines CV, DB, and MS. In the loss-function ablations, CV and MS are held fixed while conventional MSE, DB-MSE, and Score-Balanced Loss (SB) are compared.

CCC-BS measures distribution-balanced agreement and PCC measures linear association; higher values are preferred for both. Micro-MSE and Macro-MSE are error measures on the unified $[0, 10]$ score scale, so lower values are preferred. Boldface marks the best value in each comparison. These metrics express complementary evaluation objectives; improvements under balanced criteria need not coincide with improvements under sample-weighted criteria.

II. RESULTS

M3C shows a strong response to the complete methodology: BOMA is best on eight of nine CCC-BS aspects, with +DB remaining best only for utterance completeness (Table S1). The same eight-to-one pattern is observed for Macro-MSE, while the Baseline minimizes Micro-MSE on all nine aspects (Table S2). PCC again provides a contrasting view, with the Baseline leading all nine aspects (Table S3). In the controlled loss comparison, DB-MSE is best on eight CCC-BS aspects and eight Macro-MSE aspects, with SB leading utterance completeness in both cases (Tables S4 and S5).

TABLE S1
ABLATION STUDY FOR M3C. CORRELATIONS ARE MEASURED USING CCC-BS.

Configuration	Phone Score	Word Score (CCC-BS)			Utterance Score (CCC-BS)				
	Phoneme $\uparrow$	Acc. $\uparrow$	Stress $\uparrow$	Total $\uparrow$	Acc. $\uparrow$	Comp. $\uparrow$	Fluency $\uparrow$	Prosody $\uparrow$	Total $\uparrow$
Baseline	0.544	0.574	0.125	0.595	0.674	0.004	0.760	0.771	0.796
+ CV	0.462	0.549	0.157	0.559	0.653	-0.016	0.687	0.691	0.771
+ DB	0.588	0.657	0.648	0.593	0.732	0.191	0.752	0.733	0.762
+ MS	0.487	0.556	0.166	0.562	0.622	0.007	0.565	0.588	0.650
BOMA	0.641	0.674	0.702	0.610	0.786	0.056	0.794	0.785	0.828

^a The +DB and +MS configurations are evaluated on top of the CV configuration; BOMA combines CV, DB, and MS.

^b Bold values indicate the highest CCC-BS value for each aspect within this ablation.

TABLE S2
MICRO-MSE AND MACRO-MSE ABLATION STUDY FOR M3C. LOWER VALUES INDICATE BETTER PERFORMANCE.

Configuration	Phone		Word Score						Utterance Score									
	Phoneme		Acc.		Stress		Total		Acc.		Comp.		Fluency		Prosody		Total	
	Mic. $\downarrow$	Mac. $\downarrow$	Mic. $\downarrow$	Mac. $\downarrow$	Mic. $\downarrow$	Mac. $\downarrow$	Mic. $\downarrow$	Mac. $\downarrow$	Mic. $\downarrow$	Mac. $\downarrow$	Mic. $\downarrow$	Mac. $\downarrow$	Mic. $\downarrow$	Mac. $\downarrow$	Mic. $\downarrow$	Mac. $\downarrow$	Mic. $\downarrow$	Mac. $\downarrow$
Baseline	1.639	13.178	1.588	10.954	0.183	9.718	1.198	8.824	0.957	2.597	0.261	28.790	0.676	4.351	0.697	3.876	0.869	3.446
+ CV	1.946	15.530	2.646	9.472	0.798	7.383	1.986	8.648	1.024	2.849	0.282	29.911	0.864	5.244	0.901	4.996	0.957	3.773
+ DB	4.021	8.290	7.534	5.075	1.952	3.055	6.290	5.430	1.411	3.227	0.569	21.818	1.185	4.337	1.162	4.183	1.395	3.565
+ MS	2.059	14.253	3.302	8.555	1.311	6.598	2.255	8.552	1.257	3.361	0.291	28.893	1.099	7.209	1.110	6.354	1.216	5.092
BOMA	5.629	6.368	8.064	4.710	2.208	2.711	5.922	5.171	1.212	2.335	0.304	24.485	1.028	3.464	1.013	3.334	1.105	2.678

^a Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified $[0, 10]$ scale; lower values ($\downarrow$) indicate better performance.

^b The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

^c Bold values indicate the lowest error for each metric within the model-specific ablation.

TABLE S3
PCC ABLATION STUDY FOR M3C. HIGHER VALUES INDICATE BETTER PERFORMANCE.

Configuration	Phone Score	Word Score (PCC)			Utterance Score (PCC)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
Baseline	0.716	0.710	0.340	0.721	0.791	0.268	0.830	0.832	0.816
+ CV	0.653	0.649	0.276	0.662	0.763	-0.008	0.763	0.756	0.783
+ DB	0.553	0.539	0.247	0.563	0.699	0.209	0.698	0.684	0.717
+ MS	0.639	0.629	0.277	0.642	0.676	0.045	0.661	0.665	0.689
BOMA	0.541	0.550	0.244	0.569	0.738	0.169	0.746	0.740	0.764

^a The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

^b Bold values indicate the highest PCC value for each aspect within the model-specific ablation.

TABLE S4
LOSS-FUNCTION ABLATION IN TERMS OF CCC-BS FOR M3C. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

Loss	Phone Score	Word Score (CCC-BS)			Utterance Score (CCC-BS)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
MSE	0.487	0.556	0.166	0.562	0.622	0.007	0.565	0.588	0.650
DB-MSE	0.641	0.674	0.702	0.610	0.786	0.056	0.794	0.785	0.828
SB	0.453	0.458	0.157	0.449	0.192	0.105	0.121	0.127	0.132

^a MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

^b DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

^c SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

^d Bold values indicate the highest CCC-BS for each aspect within the model-specific loss ablation.

TABLE S5
LOSS-FUNCTION ABLATION IN TERMS OF MICRO-MSE AND MACRO-MSE FOR M3C. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

Loss	Phone		Word Score						Utterance Score									
	Phoneme		Acc.		Stress		Total		Acc.		Comp.		Fluency		Prosody		Total	
	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓
MSE	2.059	14.253	3.302	8.555	1.311	6.598	2.255	8.552	1.257	3.361	0.291	28.893	1.099	7.209	1.110	6.354	1.216	5.092
DB-MSE	5.629	6.368	8.064	4.710	2.208	2.711	5.922	5.171	1.212	2.335	0.304	24.485	1.028	3.464	1.013	3.334	1.105	2.678
SB	2.404	14.417	4.528	9.191	1.737	6.225	6.159	7.990	2.204	6.358	0.297	24.145	1.936	14.747	1.949	13.468	2.263	11.586

^a MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

^b DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

^c SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

^d Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified [0, 10] scale; lower values (↓) indicate better performance.

^e Bold values indicate the lowest error for each metric within the model-specific loss ablation.